# SALES AND CUSTOMER ANALYTICS SYSTEM

## Objectives :

1. Apply COUNT(), SUM(), AVG(), MIN(), MAX().
2. Generate total sales reports.
3. Find top customers based on purchase amount.
4. Identify best-selling products.
5. Perform category-wise sales analysis.

## 1. Aggregate Functions :

Aggregate functions are SQL functions used to perform calculations on a group of rows and return a single result.

## 1.1 COUNT()

Find the total number of orders:

```
SQL> SELECT COUNT(Order_ID) AS Total_Orders
  2  FROM Orders;

TOTAL_ORDERS
------------
           8
```

## 1.2 SUM()

Find the total sales amount:

```
SQL> SELECT SUM(Total_Amount) AS Total_Sales
  2  FROM Orders;

TOTAL_SALES
-----------
      12991
```

## 1.3 AVG()

Find the average order amount:

```
SQL> SELECT AVG(Total_Amount) AS Average_Order
  2  FROM Orders;

AVERAGE_ORDER
-------------
     1623.875
```

## 1.4 MIN()

Find the lowest order amount:

```
SQL> SELECT MIN(Total_Amount) AS Minimum_Order
  2  FROM Orders;

MINIMUM_ORDER
-------------
          450
```

## 1.5 MAX()

Find the highest order amount:

```
SQL> SELECT MAX(Total_Amount) AS Maximum_Order
  2  FROM Orders;

MAXIMUM_ORDER
-------------
         2598
```

## 2. Generate Total Sales Reports

A sales report can be generated using the SUM() function on the Total_Amount column.

```
SQL> SELECT
  2      COUNT(Order_ID) AS Total_Orders,
  3      SUM(Total_Amount) AS Total_Sales,
  4      AVG(Total_Amount) AS Average_Order_Value,
  5      MIN(Total_Amount) AS Minimum_Order,
  6      MAX(Total_Amount) AS Maximum_Order
  7  FROM Orders;

TOTAL_ORDERS TOTAL_SALES AVERAGE_ORDER_VALUE MINIMUM_ORDER MAXIMUM_ORDER
------------ ----------- ------------------- ------------- -------------
           8       12991            1623.875           450          2598
```

## 2.1 Customer-wise Sales Report

To find how much each customer has purchased:

```
SQL> SELECT
  2      C.ID,
  3      C.NAME,
  4      SUM(O.TOTAL_AMOUNT) AS TOTAL_PURCHASE
  5  FROM Customer C
  6  JOIN Orders O
  7      ON C.ID = O.CUSTOMER_ID
  8  GROUP BY
  9      C.ID,
 10      C.NAME
 11  ORDER BY TOTAL_PURCHASE DESC;

        ID NAME                           TOTAL_PURCHASE
---------- ------------------------------ --------------
         3 kaushik                                  4597
         1 Ramesh                                   3397
         4 Chaitali                                 1999
         2 Khilan                                   1499
         5 Hardik                                   1499
```

## 3. Find Top Customers Based on Purchase Amount

The SUM() function can be combined with GROUP BY and ORDER BY to identify customers with higher purchase amounts.

```
SQL> SELECT
  2      C.ID,
  3      C.NAME,
  4      SUM(O.TOTAL_AMOUNT) AS TOTAL_PURCHASE
  5  FROM Customer C
  6  JOIN Orders O
  7      ON C.ID = O.CUSTOMER_ID
  8  GROUP BY
  9      C.ID,
 10      C.NAME
 11  ORDER BY TOTAL_PURCHASE DESC;

        ID NAME                           TOTAL_PURCHASE
---------- ------------------------------ --------------
         3 kaushik                                  4597
         1 Ramesh                                   3397
         4 Chaitali                                 1999
         2 Khilan                                   1499
         5 Hardik                                   1499
```

The customer appearing at the top has the highest purchase amount in the generated report.

## 4. Identify Best-Selling Products

The Order_Item table contains the quantity of each product ordered.Therefore, we can use SUM(Quantity) to find the total quantity sold for each product.

```
SQL> SELECT
  2      P.PRODUCT_ID,
  3      P.PRODUCT_NAME,
  4      SUM(OD.QUANTITY) AS TOTAL_QUANTITY_SOLD
  5  FROM PRODUCT P
  6  JOIN ORDER_DETAILS OD
  7      ON P.PRODUCT_ID = OD.PRODUCT_ID
  8  GROUP BY
  9      P.PRODUCT_ID,
 10      P.PRODUCT_NAME
 11  ORDER BY TOTAL_QUANTITY_SOLD DESC;

PRODUCT_ID PRODUCT_NAME
---------- --------------------------------------------------
TOTAL_QUANTITY_SOLD
-------------------
       101 Laptop
                   3

       103 Headphones
                   2

       105 Mouse
                   2
```

```
PRODUCT_ID PRODUCT_NAME
---------- --------------------------------------------------
TOTAL_QUANTITY_SOLD
-------------------
       104 Keyboard
                   1

       106 Monitor
                   1

       102 Smartphone
                   1


6 rows selected.
```

## 4.1 Product Sales Amount

Instead of only calculating quantity, we can calculate the total sales generated by each product.

```
SQL> SELECT
  2      P.PRODUCT_ID,
  3      P.PRODUCT_NAME,
  4      SUM(OD.QUANTITY) AS TOTAL_QUANTITY_SOLD,
  5      SUM(OD.SUBTOTAL) AS TOTAL_SALES
  6  FROM PRODUCT P
  7  JOIN ORDER_DETAILS OD
  8      ON P.PRODUCT_ID = OD.PRODUCT_ID
  9  GROUP BY
 10      P.PRODUCT_ID,
 11      P.PRODUCT_NAME
 12  ORDER BY TOTAL_SALES DESC;

PRODUCT_ID PRODUCT_NAME
---------- --------------------------------------------------
TOTAL_QUANTITY_SOLD TOTAL_SALES
------------------- -----------
       103 Headphones
                   2         700

       105 Mouse
                   2         500

       104 Keyboard
                   1         450
```

```
PRODUCT_ID PRODUCT_NAME
---------- --------------------------------------------------
TOTAL_QUANTITY_SOLD TOTAL_SALES
------------------- -----------
       101 Laptop
                   3         390

       102 Smartphone
                   1         180

       106 Monitor
                   1         120


6 rows selected.
```

## 5. Category-wise Sales Analysis

Category-wise analysis helps determine the sales generated by each product category.

```
SQL> SELECT
  2      C.CATEGORY_ID,
  3      C.CATEGORY_NAME,
  4      SUM(OD.QUANTITY) AS TOTAL_QUANTITY_SOLD,
  5      SUM(OD.SUBTOTAL) AS TOTAL_SALES
  6  FROM CATEGORY C
  7  JOIN PRODUCT P
  8      ON C.CATEGORY_ID = P.CATEGORY_ID
  9  JOIN ORDER_DETAILS OD
 10      ON P.PRODUCT_ID = OD.PRODUCT_ID
 11  GROUP BY
 12      C.CATEGORY_ID,
 13      C.CATEGORY_NAME
 14  ORDER BY TOTAL_SALES DESC;

CATEGORY_ID
-----------
CATEGORY_NAME
--------------------------------------------------------------------------------
TOTAL_QUANTITY_SOLD TOTAL_SALES
------------------- -----------
          1
Mobile Accessories
                  6        1830

          3
Home Appliances
                  4         510
```